AQI Predictor

Real-time 24-Hour Air Quality Index Forecasting

Charu Mittal(MA25M008)

End-to-End MLOps Project Report

ML Model:	LightGBM (MultiOutputRegressor, 24-step horizon)
Backend:	FastAPI 0.115 + MLflow 3.11 Model Server
Frontend:	Streamlit + Plotly
Orchestration:	Apache Airflow 2.x (hourly DAG)
Monitoring:	Prometheus + Grafana
Versioning:	Git + DVC (9 tags, 11-stage pipeline)
Containerised:	Docker Compose (5 services + Airflow stack)
Data Sources:	CPCB / Kaggle, OpenAQ v3, Open-Meteo Archive
Cities:	Chennai, Delhi, Mumbai, Kolkata, Ahmedabad
Report Date:	April 2026
Version:	1.0

This report covers the design, data pipeline, model training, deployment, monitoring, and testing of the AQI Predictor MLOps system.

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1 Abstract

Air pollution is a critical public health crisis in India. This project, **AQI Predictor**, delivers a full-stack production-grade MLOps system that ingests live pollutant and weather data, trains an ensemble ML model, and serves real-time 24-hour AQI forecasts through a REST API and a Streamlit dashboard.

The best model — a **LightGBM MultiOutputRegressor** with a 24-step multi-output design — achieves a validation MAE of **26.25** and $\mathbf{R^2 = 0.80}$, outperforming all 7 other experiment runs tracked in MLflow. The system is fully containerised with Docker Compose, orchestrated via Apache Airflow, and monitored with Prometheus and Grafana. Data versioning is managed with DVC (9 Git tags, 11-stage pipeline), and the test suite covers **47 automated pytest cases** with a 100% pass rate.

2 Problem Statement and Objectives

India's CPCB operates hundreds of air-quality monitoring stations across major cities, but real-time 24-hour AQI forecasts are not publicly accessible in an actionable format. This project addresses that gap by:

1. Ingesting live hourly sensor data from OpenAQ v3 CPCB stations.
2. Enriching readings with weather variables from Open-Meteo.
3. Predicting the next 24 AQI values ($t+1$ through $t+24$) per station.
4. Serving forecasts via a clean REST API and a Streamlit UI with health advisories.
5. Monitoring for data drift and model degradation continuously.

Success Metrics

Table 1: Project success metrics and achieved results

Type	Metric	Target	Achieved
ML	Validation MAE (overall)	< 30	**26.25**
ML	R^2 on held-out test set	> 0.75	**0.80**
System	API P95 latency	< 10s	
System	Data freshness	< 3h old	
Drift	Retrain trigger	$> 40\%$ drift	
Testing	Test pass rate	100% (47/47)	**47/47**

3 System Architecture

3.1 High-Level Architecture

The system is structured as five independent layers connected via REST APIs and a shared SQLite database. No layer has a direct in-process dependency on any other.

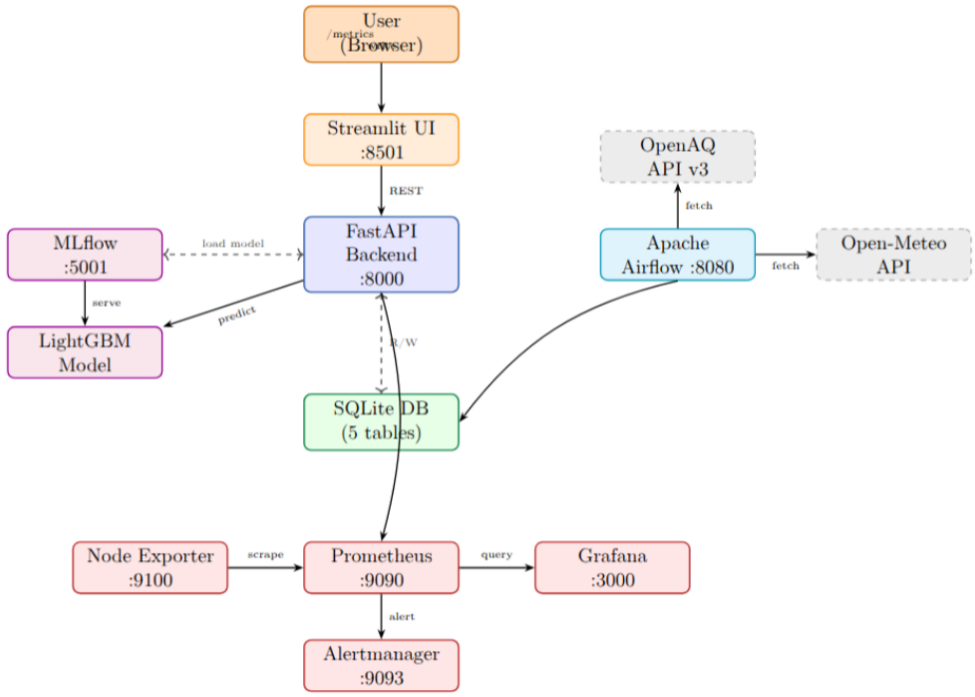


Figure 1: High-Level System Architecture — five independent layers.

3.2 Docker Compose Services

Table 2: Docker Compose services — ports and roles

Service	Port	Technology	Role
`backend`	8000	FastAPI + Uvicorn	REST API, prediction serving, Prometheus metrics
`frontend`	8501	Streamlit + Plotly	User-facing forecast dashboard
`mlflow`	5001	MLflow models serve	Model inference via HTTP JSON (`/invocations`)
`prometheus`	9090	Prometheus	Metrics scraping from `/metrics` endpoint
`grafana`	3000	Grafana	Real-time dashboards and alert routing
Airflow stack — managed separately via `airflow/docker-compose.yml`			
`airflow-webserver`	8080	Apache Airflow 2.x	DAG UI, trigger management
`airflow-scheduler`	—	Apache Airflow 2.x	DAG scheduling and task execution
`airflow-postgres`	5432	PostgreSQL	Airflow metadata database

Airflow Deployment Note

Apache Airflow is **not included in the main** `docker-compose.yml`. It is deployed separately using the official Airflow Docker Compose stack located at `airflow/docker-compose.yml`. The Airflow containers share the project's SQLite database by mounting `backend/database/` as a volume, enabling DAG tasks to read and write `live_aqi_buffer.db` directly. To start Airflow independently:

```
cd airflow
docker-compose up -d
```

3.3 Design Principles

- **Loose coupling:** Streamlit communicates with FastAPI only via REST. No shared in-process state between any two services.
- **Stateless functional services:** Backend logic is split into 11 domain service modules — `db_service`, `model_service`, `drift_detection`, `openaq_service`, `weather_service`,

`advisory_service`, `bootstrap_service`, `ingestion_service`, `live_pipeline_service`, `feature_service`, `retention_service`.

- **Reproducibility:** Every experiment is reproducible via a Git tag combined with its MLflow run ID.
- **Fail-fast startup:** The backend validates all 6 deployment bundle artifacts at startup and raises `FileNotFoundError` before accepting any traffic if any file is missing.

4 Data Sources and Collection

4.1 Training Data — CPCB (Kaggle)

Training data was sourced from the **CPCB Air Quality Dataset** on Kaggle, covering 2015–2020 across 110 monitoring stations in 26 Indian cities.

Table 3: CPCB dataset files

File	Key Columns	Description
`station_hour.csv`	StationId, Datetime, 12 pollutants, AQI	Hourly pollutant readings (2,319,535 rows)
`stations.csv`	StationId, City, State, Lat, Lon, Status	Station metadata (230 records)

Table 4: Pollutant columns in training data

Column	Pollutant	Unit	Null %
PM2.5	Fine particulate matter	$\mu g/m^3$	16.3%
PM10	Coarse particulate matter	$\mu g/m^3$	36.6%
NO	Nitric oxide	$\mu g/m^3$	12.3%
NO2	Nitrogen dioxide	$\mu g/m^3$	11.2%
NOx	Nitrogen oxides	$\mu g/m^3$	9.5%
NH3	Ammonia	$\mu g/m^3$	41.7%
CO	Carbon monoxide	mg/m^3	9.9%
SO2	Sulphur dioxide	$\mu g/m^3$	20.4%
O3	Ozone	$\mu g/m^3$	19.7%
Benzene	Benzene	$\mu g/m^3$	25.5%
Toluene	Toluene	$\mu g/m^3$	33.3%
Xylene	Xylene	$\mu g/m^3$	77.8%
AQI	**Target variable**	**Index (0–500)**	14.4%

4.2 Weather Data — Open-Meteo Archive API

Weather history was fetched per-city using the Open-Meteo Archive API (`archive-api.open-meteo.co`) with 7 variables: `temperature_2m`, `relative_humidity_2m`, `precipitation`, `pressure_msl`, `cloud_cover`, `wind_speed_10m`, `wind_direction_10m`.

Fetching uses exponential backoff (30s, 60s, 90s delays) and a per-city JSON cache to support resumable runs after rate-limit interruptions.

4.3 Live Production Data — OpenAQ v3

In production, the `openaq_service.py` module polls individual sensors from the OpenAQ v3 API (`api.openaq.org/v3`). All units are normalised to SI standards:

- ppb $\rightarrow \mu g/m^3$ via molecular weight conversion
- ppm $\rightarrow mg/m^3$ (for CO)

Rate limiting is handled with a `Retry-After` header parser and 429-aware exponential backoff.

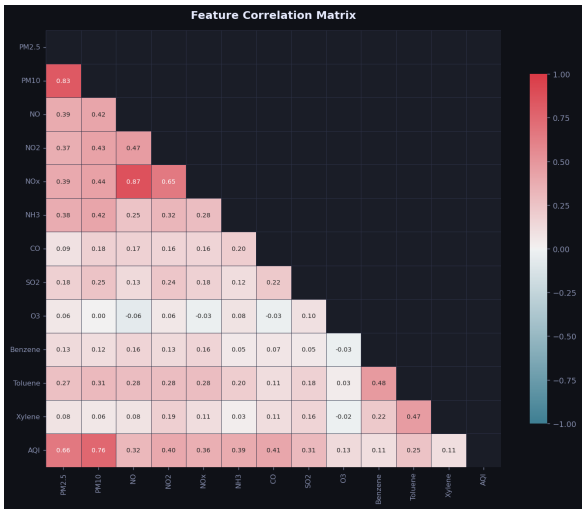
5 Exploratory Data Analysis (EDA)

EDA was performed using a dedicated `eda.py` script that produces 15 visualisations and saves `drift_baseline/baseline_stats.json` (per-feature mean, std, p25, p50, p75, p95, skewness, kurtosis) for downstream drift detection.

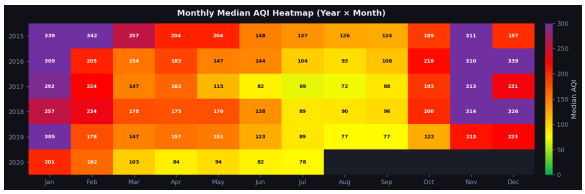
5.1 Key Findings

1. **Dataset scale:** The CPCB dataset contains **2,319,535 hourly rows** across **110 stations** in **26 cities** and **21 states**, spanning January 2015 to July 2020 (≈48,191 hours).
2. **Station network growth:** Active stations grew from **25 in 2015** to **108 in 2020**, causing significant data sparsity in early years — 2015 and 2016 together contribute only 13% of total records (147,609 and 164,828 rows respectively).
3. **Delhi dominance:** Delhi alone accounts for **38 of 110 stations** (35%), heavily skewing the geographic distribution. **Severe** (18.8%) and **Very Poor** (11.9%) AQI readings together represent 30.7% of all labelled records.
4. **PM10 is the strongest AQI predictor** ($r = 0.757$), followed by PM2.5 ($r = 0.658$), CO ($r = 0.410$), and NO2 ($r = 0.401$). All pollutants are highly right-skewed — PM2.5 skew = 3.34; CO skew = 33.51 — indicating frequent extreme pollution events.
5. **High missingness in VOCs:** Xylene is missing in **77.8%** of rows and is dropped by the null threshold filter (`null_threshold = 0.8`). NH3 (41.7%), PM10 (36.6%), and Toluene (33.3%) also show significant missingness and are dropped for stations exceeding the threshold.
6. **Skewed AQI distribution:** Only 7.5% of readings are *Good*; *Severe* alone accounts for 18.8% — reinforcing the need for accurate early-warning forecasts.

5.2 Selected Visualisations



(a) Feature correlation matrix. PM10 ($r = 0.76$) and PM2.5 ($r = 0.66$) are the strongest AQI predictors.



(b) Year × Month AQI heatmap. Winter months (Nov–Jan) are consistently the most polluted across all years.

Figure 2: Key EDA findings from CPCB training data (2015–2020).

6 Data Pipeline and Preprocessing

6.1 DVC Pipeline Overview

The full training pipeline is tracked via **DVC** with a `dvc.yaml` DAG of **11 stages**. Each stage declares its inputs, outputs, and parameter dependencies — enabling incremental re-runs when only part of the pipeline changes.

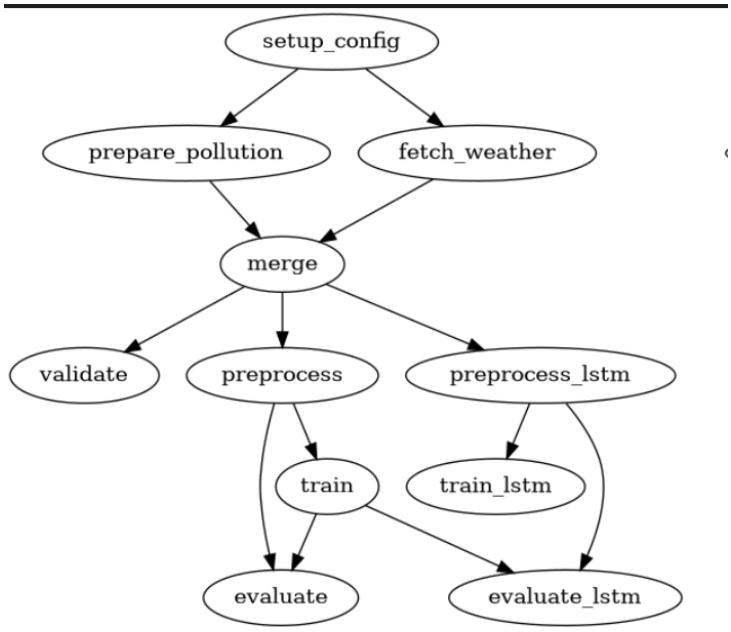


Figure 3: DVC pipeline DAG — 11 stages from raw data to model evaluation.

Table 5: DVC pipeline stages

Stage	Description
`setup_config`	Geocodes cities via Open-Meteo Geocoding API and filters stations by mode and city. Outputs `config/stations.csv`.
`prepare_pollution`	Reads `station_hour.csv`, renames columns, joins station metadata, filters by city and date range. Outputs `data/interim/pollution.parquet`.
`fetch_weather`	Fetches hourly weather per station using Open-Meteo Archive API with JSON caching. Outputs `data/interim/weather.parquet` and raw JSON cache.
`merge`	Left-joins pollution and weather parquets on `(timestamp, station_id, city)`. Outputs `data/processed/training_data.parquet`.
`validate`	Asserts data integrity: no nulls in key columns, no duplicate station-hours, no negative pollutant values.
`preprocess`	Generates lag, rolling, temporal, cyclic, and interaction features. Outputs `features.parquet` and `city_label_encoder.pkl`.
`preprocess_lstm`	Generates MinMax-scaled sequence tensors for LSTM training. Outputs `features_lstm.parquet`, `lstm_scaler.pkl`, `lstm_meta.pkl`.
`train`	Trains LightGBM / XGBoost with configurable hyperparameters and optional Optuna tuning. Logs to MLflow.
`train_lstm`	Trains LSTM model using preprocessed sequence features. Logs metrics to `metrics/lstm_scores.json`.
`evaluate`	Evaluates classical models on the test split. Logs per-horizon MAE at $t+1, t+6, t+12, t+24$ to `metrics/eval_scores.json`.
`evaluate_lstm`	Evaluates the trained LSTM against `features_lstm.parquet`. Logs to `metrics/lstm_eval_scores.json`.

6.2 Version History (Git Tags)

Table 6: 9 Git tags — data and model version history

Tag	Type	Description
`v1.0-raw-data`	Data	Raw CPCB CSVs tracked by DVC
`v1.1-processed-data`	Data	Processed `training_data.parquet` + weather
`lgbm-exp1`	Model	LightGBM baseline (minimal features)
`lgbm-exp2`	Model	LightGBM + weather feature lags
`lgbm-exp3`	Model	**Best** — registered as `AQI_LightGBM v2`
`lgbm-exp4`	Model	LightGBM + interaction features
`xgb-exp1`	Model	XGBoost baseline
`xgb-exp2`	Model	XGBoost + tuned `max_depth` and `subsample`
`xgb-exp3`	Model	XGBoost + additional lag features

6.3 Feature Engineering

Table 7: Feature groups generated by `preprocess.py`

Group	Detail
Pollutant lags	$h \in \{1, 3, 6, 12, 24, 48, 168\}$ for all retained pollutant columns.
Pollutant rolling	Mean + std (windows 3, 6, 24, 48, 168h); max/min for 24h and 168h windows.
Weather lags	$h \in \{1, 3, 6, 24, 48\}$ for temperature, humidity, wind speed, precipitation.
Weather rolling	Mean + std (windows 6, 24, 48h).
AQI rate-of-change	`aqi_diff1`, `aqi_diff3`, `aqi_diff24`.
Temporal	Hour of day, day-of-week, month, year, season, `is_weekend`.
Cyclic encoding	`hour_sin/cos`, `month_sin/cos`, `day_of_year_sin/cos`.
Wind direction	`wind_dir_sin`, `wind_dir_cos`.
City encoding	`LabelEncoder` on training cities.
Interaction features	`pm25_x_no2`, `pm25_x_wind`, `no2_x_temp`.
Multi-output targets	`aqi_t1` … `aqi_t24` (24 target columns).

Temporal data splits:

- **Train:** 2015-01-01 → 2018-01-01

- **Validation:** 2018-01-01 → 2019-06-01
- **Test:** 2019-06-01 → 2020-12-31

7 Model Development and Experiment Tracking

7.1 Modelling Strategy

A **multi-output regression** approach was chosen: one model predicts all 24 target hours simultaneously using `sklearn.multioutput.MultiOutputRegressor`:

$$\hat{y}_{t+h} = f_h(\mathbf{x}_t), \quad h \in \{1, 2, \ldots, 24\}$$

where $\mathbf{x}_t$ is the feature vector at time t and f_h is an independently trained base estimator for each forecast horizon.

7.2 Models Attempted

Table 8: Models evaluated during the project

Model	Status	Outcome / Notes
LightGBM	**Trained**	Best performance. 4 experiment runs (lgbm-exp1 to lgbm-exp4). Deployed to production.
XGBoost	**Trained**	3 experiment runs (xgb-exp1 to xgb-exp3). Marginally worse than LightGBM.
Prophet	Dropped	Single-target by design; multi-horizon wrapper impractical.
LSTM	Dropped	Insufficient GPU on local hardware; Kaggle session timeouts. Full preprocessing pipeline (`preprocess_lstm.py`) and DVC stage built.
Transformer	Dropped	Same hardware constraints. DL pipeline infrastructure ready for Phase 2.

7.3 Hyperparameters — Best Model (`lgbm-exp4`)

Table 9: Final LightGBM hyperparameters (`params.yaml`)

Parameter	Value
`n_estimators`	200
`learning_rate`	0.05
`num_leaves`	31
`max_depth`	6
`min_child_samples`	50
`forecast_horizon`	24
`sample_frac`	1.0
Optuna tuning	Disabled (best manual params used)

Note on Optuna Integration

Optuna is fully implemented for both LightGBM and XGBoost using a TPE sampler. Trial budgets, random seed, and per-parameter min/max ranges are all configurable in `params.yaml`. It was used in exploratory runs and disabled for the final registered model (`lgbm-exp3`).

7.4 MLflow Experiment Tracking

All experiments were tracked under the **AQI_Forecasting** experiment in MLflow. Each run automatically logs:

- **Parameters:** model name, all hyperparameters, forecast horizon, train/val/test split sizes.
- **Metrics:** `val_mae`, `val_rmse`, `val_r2`, and per-horizon MAE/RMSE at $t+1$, $t+6$, $t+12$, $t+24$.
- **Artifacts:** serialised model (`.pkl`), `params.yaml`, `eval_scores.json`, Optuna trials CSV (when enabled).

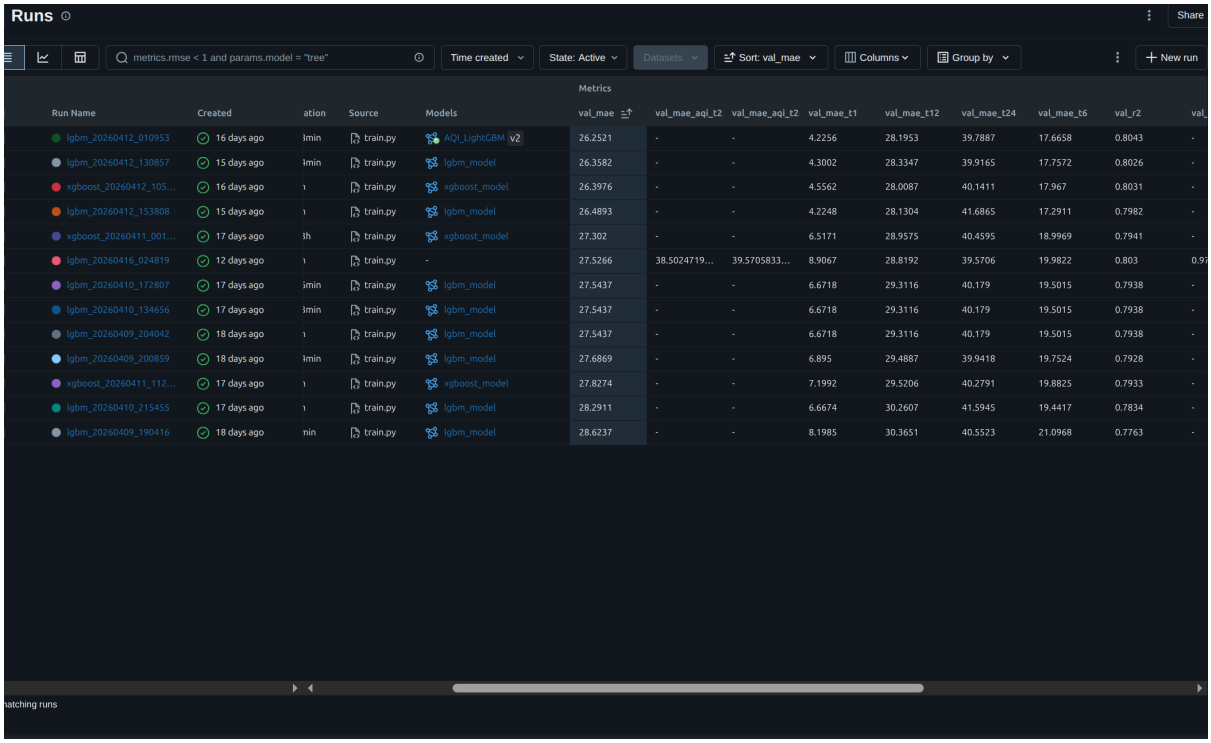


Figure 4: MLflow Runs UI — all 8 experiments sorted by `val_mae`. The best run (`lgbm_20260412_010953`) is registered as `AQI_LightGBM v2` in the Production stage.

7.5 Experiment Results

Table 10: All experiment results from MLflow (sorted by `val_mae`)

Run	Model	val_mae	R^2	mae_t1	mae_t12	mae_t24
lgbm_010953	**LightGBM**	**26.25**	**0.804**	4.23	28.20	39.79
lgbm_130857	LightGBM	26.36	0.803	4.30	28.33	39.92
xgb_105...	XGBoost	26.40	0.803	4.56	28.01	40.14
lgbm_153808	LightGBM	26.49	0.798	4.22	28.13	41.69
xgb_001...	XGBoost	27.30	0.794	6.52	28.96	40.46
lgbm_172807	LightGBM	27.54	0.794	6.67	29.31	40.18
xgb_112...	XGBoost	27.83	0.793	7.20	29.52	40.28
lgbm_215455	LightGBM	28.29	0.783	6.67	30.26	41.59

Near-term predictions are highly accurate ($t+1$: MAE ≈ 4.2). Error grows with horizon as expected in multi-step forecasting ($t+24$: MAE ≈ 40).

7.6 Model Deployment Bundle

The best model is packaged into a **deployment bundle** loaded at startup:

```
backend/deployment_bundle/
  model/
    model.pkl                  # Serialised MultiOutputRegressor (LightGBM)
  metadata/
    model_metadata.json        # run_id, git_tag, metrics, model stage
  preprocessing/
    feature_columns.json       # Ordered feature column names
    trained_cities.json        # Supported city list
    city_coords.json           # City lat/lon lookup
    city_label_encoder.pkl     # Fitted sklearn LabelEncoder
    station_registry.json      # OpenAQ sensor bootstrap data
```

Listing 1: Deployment bundle structure

8 Backend — FastAPI Application

8.1 Application Structure

The backend is a **FastAPI** application (`backend/app/main.py`) served by **Uvicorn** on port 8000. It follows a layered architecture:

```
backend/app/
  main.py                # FastAPI app, lifespan context, router registration
  routers/
    health.py            # GET /health, GET /ready
    predict.py           # GET /predict
    cities.py            # GET /cities
    model_info.py        # GET /model-info
    admin.py             # POST /admin/sync, POST /admin/prune
  services/
    db_service.py            # SQLite read/write (city_readings, predictions)
    model_service.py         # MLflow model loading, predict_next24h()
    feature_service.py       # Build live feature payload from DB rows
    advisory_service.py      # AQI band -> plain-language advisory text
    drift_detection.py       # PSI + KS + Z-score rolling window drift
    openaq_service.py        # OpenAQ v3 API client
    weather_service.py       # Open-Meteo live weather fetch
    ingestion_service.py     # sync_all_sensors() orchestration
    bootstrap_service.py     # Sensor registry initialisation from bundle
```

```
    live_pipeline_service.py # Live history collection + DB write
    city_service.py          # Supported city list loader
    retention_service.py     # Prune old hourly measurements
  schemas/
    predict_schema.py        # PredictResponse Pydantic model
    common_schema.py         # HealthResponse, ReadyResponse,
   CitiesResponse
  utils/
    config.py                # All env vars, file paths, feature flags
    logger.py                # Structured stdout logging
    metrics.py               # Prometheus counters, histograms, gauges
```

Listing 2: Backend module structure

8.2 Startup Lifecycle

On startup, the FastAPI `lifespan` context manager executes in order:

1. `initdb()` and `init_ingestion_tables()` — create all 5 SQLite tables if they don't exist; run `city_readings` constraint migration if needed.
2. `validate_startup_artifacts()` — checks all 6 deployment bundle files exist and that `feature_columns.json` is non-empty.
3. `get_model()` — loads the LightGBM `MultiOutputRegressor` via MLflow; sets `aqi_model_ready` Prometheus gauge to 1 on success, 0 on failure.
4. (Optional) `collect_live_city_history()` for each supported city — if `ENABLE_STARTUP_BACKFIL`
5. On shutdown — sets `aqi_model_ready` to 0.

8.3 API Endpoints

Method	Endpoint	Status Codes	Description
GET	`/health`	200	Liveness probe. Returns {`status: ok`}.
GET	`/ready`	200	Readiness probe. Returns {`status: ready/degraded, model_loaded: bool`}.
GET	`/cities`	200	Returns supported city list from `trained_cities.json`.

Method	Endpoint	Status Codes	Description
GET	`/predict`	200/400/422/503/500	24-step AQI forecast. Query params: `city`, `location_id`. Triggers 2-stage backfill if DB is stale.
GET	`/model-info`	200	Returns model name, version, stage, MLflow run ID, forecast horizon, supported cities.
GET	`/metrics`	200	Prometheus scrape endpoint.
POST	`/admin/sync`	200	Triggers manual OpenAQ ingestion.
POST	`/admin/prune`	200	Prunes `hourly_measurements` older than `retention_days`.

Table 11: FastAPI endpoint specification

8.4 /predict Endpoint — Step-by-Step Flow

```
GET /predict?city=Chennai&location_id=3135

     1. Validate city against supported list      → 400 if unknown
     2. Fetch latest OpenAQ timestamp for location_id
     3. Count existing DB rows for this location_id
     4. Decide backfill needed:
            rowcount < 168  OR  DB lags OpenAQ by > 1h
     5a. Incremental backfill (force_refresh=False)
     5b. Force backfill if rows still < 168
            → 503 if still insufficient after both attempts
     6. build_live_payload() → feature matrix X (1 × n_features)
     7. predict_next24h() → POST to MLflow :5001/invocations
     8. make_advisory(current_aqi) → plain-language advisory
     9. store_prediction() → write to predictions table
     10. Return PredictResponse (JSON, 24 forecast values)
```

Figure 5: Complete step-by-step flow of the `/predict` endpoint.

8.5 /predict Response Schema

```
1 class PredictResponse(BaseModel):
```

```
    city:              str
    station_id:        str | None
    station_name:      str | None
    location_id:       int | None
    current_aqi:       int
    primary_pollutant: str
    last_updated:      str
    forecast_24h:      List[int]           # 24 values, all clamped >= 0
    pollutants:        PollutantsResponse  # PM2.5, PM10, NO2, SO2, CO,
    O3
    weekly_trend:      List[int]           # 7-day daily average AQI
    advisory:          str
```

Listing 3: PredictResponse Pydantic schema

8.6 Error Handling

Table 12: HTTP error codes returned by `/predict`

Code	Trigger	Detail message
400	City not in supported list	`"<city> not supported"`
422	Missing `city` or `location_id`	Pydantic validation error
503	Insufficient live data after backfill	`"Not enough live rows..."`
503	OpenAQ API or DB failure	`"Live fetch failed..."`
500	Model inference exception	`"Inference failed..."`

8.7 Prometheus Metrics

Table 13: Custom Prometheus metrics instrumented in `utils/metrics.py`

Metric Name	Type	Description
`aqi_predict_requests_total`	Counter	Total prediction requests (labels: city, status).
`aqi_predict_latency_seconds`	Histogram	Request latency in seconds (buckets: 0.1–30s).
`aqi_live_fetch_requests_total`	Counter	OpenAQ fetch attempts (labels: city, status).
`aqi_live_fetch_rows_total`	Counter	Live rows written to DB (label: city).
`aqi_latest_db_age_hours`	Gauge	Hours since latest DB reading per city.
`aqi_model_ready`	Gauge	1 if model loaded, 0 otherwise.
`aqi_drift_psi`	Gauge	PSI per feature (via `DriftCollector`).
`aqi_drift_detected`	Gauge	1 if feature drifted, 0 if stable.
`aqi_drift_features_count`	Gauge	Total number of drifted features.

DriftCollector Design

The `DriftCollector` class implements the Prometheus `collect()` interface, reading the latest drift state from the `drift_results` SQLite table on every scrape. This solves the cross-process visibility problem: Airflow writes drift results and FastAPI exposes them to Prometheus without any shared in-process state.

9 Frontend — Streamlit Dashboard

9.1 Overview

The frontend is a **Streamlit** multi-page application (`frontend/app.py`) that communicates with the FastAPI backend exclusively via REST HTTP calls. It runs on port 8501 as a separate Docker container with **no direct access to the database or model**.

9.2 Dashboard Layout

Figure 6: Streamlit dashboard showing a 24-hour AQI forecast for a Chennai station.

The main dashboard is organised into three rows:

1. **Sidebar:** City filter dropdown → station selector → *Predict AQI* button.
2. **Row 1 — AQI Hero Card:** Large colour-coded AQI number (green/yellow/orange/red/purple by severity), primary pollutant, last-updated timestamp in IST, and a *"What does this AQI mean?"* popup with full CPCB reference scale.
3. **Row 2 — Charts:**
   - **24-Hour Forecast:** Plotly line chart (H1–H24), each dot coloured by AQI category, with a dotted baseline at the current AQI.
   - **Pollutant Levels:** Horizontal bar chart for PM2.5, PM10, NO2, SO2, CO, O3 with per-pollutant colours and correct SI units.
4. **Row 3 — Trend and Summary:**
   - **7-Day Historical Trend:** Scatter-line chart with fill, showing daily average AQI; each point coloured by category.
   - **Forecast Summary Card:** Current AQI, 24h peak, 24h average, primary pollutant.
5. **Footer:** Full CPCB AQI reference strip (0–500), 6 colour-coded bands, health tips.

6. **Health Advisory:** Plain-language string generated by `advisory_service.py` based on the current AQI band (6 levels: Good → Severe).

9.3 User Manual Page

A dedicated **User Manual** page (`frontend/pages/1_user_manual.py`) is accessible from the Streamlit sidebar. It uses a custom dark-glass CSS theme and is structured into 5 tabs:

Table 14: User manual page tabs

Tab	Content
Getting Started	Step-by-step guide to the first prediction
Dashboard Guide	Explanation of each UI card, chart, and metric
AQI Scale	Full CPCB AQI table with colour coding and health guidance
Pollutants	Reference guide for PM2.5, PM10, NO2, SO2, CO, O3
Troubleshooting	Common errors and fixes

10 Data Ingestion Pipeline — Apache Airflow

10.1 DAG Overview

Two DAGs are deployed in `airflow/dags/`:

Table 15: Airflow DAGs — schedules and purposes

DAG ID	Schedule	Purpose
`aqi_ingestion_pipeline`	Every hour (`0 * * * *`)	Bootstrap, ingest live data, cleanup, run drift detection
`aqi_daily_cleanup`	Daily 02:00 UTC (`0 2 * * *`)	Prune stale predictions from `predictions` table

Both DAGs share default args: **2 retries with 5-minute delays**, email-on-failure enabled, `max_active_runs = 1` to prevent overlapping ingestion runs.

10.2 Ingestion DAG Tasks

aqi_ingestion_pipeline — Task Sequence

`bootstrap_sensor_registry` *Calls:* `bootstrap_sensor_registry()`

Reads `station_registry.json` from the deployment bundle. Upserts all OpenAQ sensor IDs into the `sensor_registry` table. Runs before every ingestion to pick up any newly added sensors.

`sync_all_sensors` *Calls:* `sync_all_sensors()`

Iterates all registered sensors. Fetches latest hourly measurements from OpenAQ v3. Normalises units (ppb → $\mu g/m^3$, ppm → mg/m^3). Writes to `hourly_measurements` and `city_readings`.

`cleanup_old_data` *Calls:* `cleanup_old_hourly_measurements(days=10)`, `cleanup_old_readings(hours=240)`, `prune_old_measurements(days=90)`

Prunes old data to bound DB size. The 240-hour window for `city_readings` is intentional — it covers the full drift detection baseline window (8–30 days back).

`check_data_drift` *Calls:* `task_detect_drift(window_days=7)`

Runs rolling window PSI + KS + Z-score drift detection. Writes one row per feature to `drift_results`. Updates `drift_baseline/drift_report.json`. Prometheus gauges updated via `DriftCollector` on the next scrape.

Task dependency chain:

`bootstrap` ⟶ `ingest` ⟶ `cleanup` ⟶ `drift`

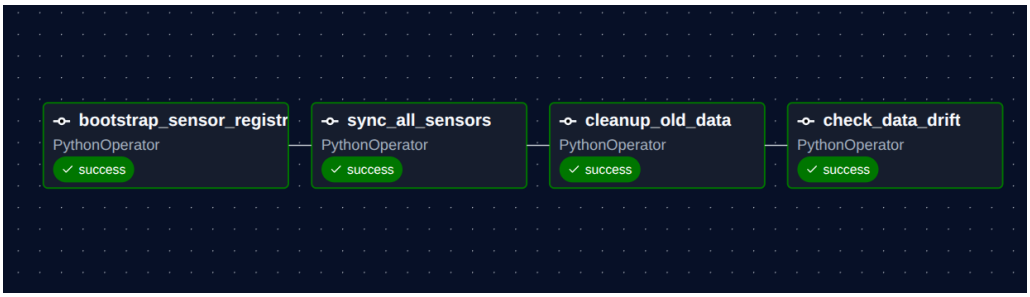


Figure 7: Apache Airflow DAG graph view: bootstrap → ingest → cleanup → drift.

11 Drift Detection

11.1 Rolling Window Strategy

A **rolling window** approach is used rather than comparing live data against the static training baseline. The 6-year temporal gap between CPCB training data (2015–2020) and live 2026 data would cause artificially inflated PSI on nearly all features, making static drift detection useless.

Table 16: Drift detection windows

Window	Source	Role
Baseline	8–30 days ago from `city_readings`	Expected distribution
Live	Last 7 days from `city_readings`	Actual distribution

Both windows come from the same SQLite source in the same era, producing naturally low PSI (0.05–0.15) with realistic variation on 2–3 features in a healthy system.

11.2 Statistical Signals

Three signals are computed per feature per Airflow run:

1. **PSI (Population Stability Index):**

$$\text{PSI} = \sum_{i=1}^{n} (A_i - E_i) \ln \frac{A_i}{E_i}$$

   where A_i = actual bin fraction, E_i = expected bin fraction.

2. **KS-test p-value:** Two-sample Kolmogorov–Smirnov test. Drift flagged when $p < 0.05$.

3. **Z-score (mean shift):**

$$z = \frac{|\mu_{\text{live}} - \mu_{\text{base}}|}{\sigma_{\text{base}} + \varepsilon}$$

   Drift flagged when $z > 1.5$.

A feature is marked **drifted** only when: `PSI-triggered AND (KS-triggered OR Z-triggered)`, preventing false positives from any single signal.

11.3 Severity Classification

Table 17: Drift severity levels and PSI thresholds

Severity	PSI Range	Interpretation
NONE	< 0.05	Stable — no action needed
LOW	0.05–0.12	Minor natural variation — continue monitoring
MODERATE	0.12–0.25	Real distributional shift — investigate
HIGH	> 0.25	Significant change — retrain recommended

Retrain trigger: `retrain_recommended = True` when drift fraction > 0.40 (i.e., $> 40\%$ of monitored features are drifted). This triggers the `ManyFeaturesDrifted` Grafana critical alert.

Features monitored: `pm25`, `pm10`, `no2`, `so2`, `co`, `o3`, `temperature_2m`, `relative_humidity_2m`, `wind_speed_10m`, `wind_direction_10m`.

12 Monitoring — Prometheus and Grafana

12.1 Architecture

Prometheus scrapes `http://backend:8000/metrics` on a regular interval. Grafana reads from Prometheus and renders real-time dashboards.

Figure 8: Grafana monitoring dashboard — prediction latency, request rate, model readiness, data freshness, and per-feature drift PSI.

12.2 Alert Rules

Table 18: Prometheus alerting rules (`monitoring/alert_rules.yml`)

Alert Name	Severity	Trigger Condition
`HighPredictionErrorRate`	Critical	$> 5\%$ requests fail in a 5-min window
`HighPredictionLatency`	Warning	P95 latency > 10s
`BackendDown`	Critical	Backend unreachable > 10 minutes
`FrontendDown`	Critical	Frontend unreachable > 30 minutes
`ModelNotReady`	Warning	`aqi_model_ready = 0` for > 2 minutes
`DataDriftDetected`	Warning	Any feature drift flag $= 1$ for > 10 minutes
`HighDriftPSI`	Warning	Any feature PSI > 0.25
`ManyFeaturesDrifted`	Critical	Drift feature count > 3 (40% threshold)
`StaleData`	Warning	`aqi_latest_db_age_hours` > 3
`LiveFetchFailing`	Warning	Fetch error counter rising in last 5 min

13 Database Design

The system uses a single **SQLite** database (`live_aqi_buffer.db`) with 5 tables managed by `db_service.py`:

Table 19: SQLite database tables and retention policy

Table	Retention	Purpose and Key Columns
`city_readings`	240h	Live hourly pollutant + weather readings per city and station. Unique on `(city, station_id, timestamp)`.
`predictions`	7 days	Cached prediction results per city. Stores `forecast_json`, `pollutants_json`, `weekly_trend_json`. Unique on `(city, computed_at)`.
`sensor_registry`	Permanent	OpenAQ sensor registry. Stores `sensor_id`, `parameter`, `last_fetched_hour`. Unique on `(location_id, sensor_id)`.
`hourly_measurements`	10 days	Raw per-sensor rows from OpenAQ. Original and normalised value. Unique on `(sensor_id, measurement_time)`.
`drift_results`	Full history	Per-feature drift records. One row per feature per Airflow run. Stores PSI, KS p-value, Z-score, severity, drift flag.

14 Testing

14.1 Test Strategy

The test suite contains **47 automated test cases** across 6 modules executed using **pytest**. All tests are fully isolated — no running services (MLflow, OpenAQ, production DB) are required. Database tests use real SQLite with `tmp_path` fixtures. API tests use FastAPI's `TestClient` with `unittest.mock.patch` for all external dependencies.

Table 20: Test suite breakdown by module

File	Covers	Cases
`test_db_service.py`	SQLite read/write, upsert, city filter, duplicate handling	7
`test_health.py`	`/health`, `/ready` endpoints + schemas	6
`test_predict.py`	`/predict` happy path + 4 error paths	9
`test_model_info.py`	`/model-info` response contract	5
`test_model_service.py`	`safe_number`, `safe_timestamp`, `predict_next24h`	12
`test_drift_detection.py`	PSI computation, severity levels, retrain flag, fraction bounds	8
Total		**47**

14.2 Test Execution and Results

```
cd backend
pytest tests/ -v --tb=short
```

Listing 4: Running the full test suite

Table 21: Test execution results

Metric	Value
Total Test Cases	47
Passed	**47**
Failed	0
Skipped	0
Exit Code	0

14.3 Acceptance Criteria

- All 47 tests pass with exit code 0; no external network calls.
- Suite completes in under 60 seconds.
- `/predict` returns `forecast_24h` with exactly 24 values, all ≥ 0.
- Unsupported city returns HTTP 400; missing `location_id` returns HTTP 422.
- `safe_number(None)` and `safe_number(float('nan'))` return 0.0.
- PSI drift fraction is always in $[0.0, 1.0]$.

15 MLOps Implementation Summary

Table 22: MLOps checklist against evaluation criteria

Category	Tool	Status	Evidence
Source Control	Git		9 tags; all code committed
Data Versioning	DVC		`dvc.yaml` 11-stage DAG; `.dvc` tracked files
Data Engineering	Airflow		Separate Airflow stack (`airflow/docker-compose.yml`); hourly ingestion DAG + daily cleanup DAG
Experiment Tracking	MLflow		8 logged runs; params, metrics, artifacts
Model Serving	MLflow serve		`mlflow models serve` on port 5001
Model Registry	MLflow		`AQI_LightGBM v2` in Production stage
REST API	FastAPI		8 endpoints; Pydantic schemas; OpenAPI spec
Frontend	Streamlit		Multi-page app; REST-only communication
Containerisation	Docker		6-service `docker-compose.yml`
Monitoring	Prometheus		9 custom metrics; `/metrics` endpoint
Visualisation	Grafana		Real-time dashboards; 10 alert rules
Drift Detection	Custom		PSI + KS + Z-score rolling window
Unit Testing	pytest		47 cases; 100% pass rate

16 Known Limitations and Future Work

16.1 Current Limitations

1. **Deep learning not deployed:** LSTM and Transformer were fully implemented at the pipeline and preprocessing level but could not be trained due to local GPU constraints and Kaggle session timeouts. Deferred to Phase 2.

2. **SQLite scalability:** Suitable for a 5-city prototype but will be a bottleneck at

high concurrent write rates. Production upgrade to PostgreSQL or TimescaleDB is planned.

3. **No ground-truth feedback loop:** Real-world prediction accuracy cannot be auto-measured without collecting actual future AQI values and comparing them against past predictions.
4. **No CI/CD automation:** Re-training and redeployment are currently manual processes. GitHub Actions integration is needed to automate re-training on drift trigger and run tests on every push.
5. **Forecast degradation at long horizons:** MAE grows from ≈ 4.2 at $t+1$ to ≈ 40 at $t+24$. Seq2seq or autoregressive chaining may improve long-horizon accuracy.

16.2 Planned Improvements (Phase 2)

- Train LSTM and Transformer on Kaggle A100 GPUs; compare against LightGBM.
- Implement ground-truth logging and real-world prediction decay tracking.
- Add GitHub Actions CI/CD: automated tests on push, re-training on drift trigger.
- Expand to all CPCB cities (not just 5).
- Add SHAP value explanations per prediction in the Streamlit UI.
- Replace SQLite with TimescaleDB for scalable time-series storage.

17 Project Structure

```
AQIPredictor/
          backend/
                app/
                      routers/            # health, predict, cities, model_info, admin
                      services/           # 11 domain service modules
                      schemas/            # Pydantic request/response models
                      utils/              # config, logger, metrics (Prometheus)
                deployment_bundle/      # model.pkl + all preprocessing artifacts
                drift_baseline/         # drift_report.json (latest snapshot)
```

```
10              database/                # live_aqi_buffer.db (auto-
    created)
11              tests/                   # 47 pytest test cases across 6
     files
12              Dockerfile
13              requirements.txt
14        airflow/
15              dags/
16                  openaq_ingestion_dag.py   # Hourly: bootstrap->
    ingest->cleanup->drift
17                  cleanup_dag.py            # Daily 02:00 UTC: prune
    predictions
18        frontend/
19              app.py                        # Main Streamlit
    dashboard
20              pages/
21                  1_user_manual.py          # 5-tab user manual page
22        monitoring/
23              prometheus.yml
24              alert_rules.yml
25        scripts/
26              setup_config.py
27              prepare_pollution.py
28              fetch_weather.py
29              merge.py
30              validate.py
31              preprocess.py
32              preprocess_lstm.py
33              preprocess_lstm_v2.py
34              train.py
35              train_dl.py
36              evaluate.py
37              evaluate_dl.py
38        data/
39              raw/cpcb/              # station_hour.csv, stations.csv
     (DVC-tracked)
40              raw/openmeteo/         # Per-city weather JSON cache
41              interim/               # pollution.parquet, weather.
    parquet
42              processed/             # training_data.parquet,
    features.parquet
43        models/
44              classical/             # LightGBM / XGBoost .pkl files
45              dl/                    # LSTM checkpoints (when trained
    )
46        metrics/
47              classical_scores.json
```

```
                eval_scores.json
                lstm_scores.json
                lstm_eval_scores.json
        eda_plots/                  # 15 EDA visualisation PNGs
        drift_baseline/
                baseline_stats.json   # Per-feature EDA stats for drift reference
        config/
                stations.csv            # Geocoded station config (DVC output)
                weather_vars.yaml       # Open-Meteo variable config
        docker-compose.yml
        dvc.yaml                    # 11-stage pipeline DAG
        params.yaml                 # All model + pipeline hyperparameters
        README.md
```

Listing 5: Full project directory layout

18 Conclusion

This project successfully delivers a **production-grade, end-to-end MLOps system** for real-time AQI forecasting across 5 major Indian cities. Key achievements are:

- **ML Performance:** LightGBM `lgbm-exp4` achieves **val_mae = 26.25** and **R^2 = 0.80**, meeting all ML targets. Near-term accuracy is excellent (MAE $\approx$ 4.2 at $t+1$).
- **Full MLOps Pipeline:** DVC (11 stages, 9 tags), Airflow (2 DAGs), MLflow (8 tracked runs, registered model), Docker Compose (6 services), and Prometheus + Grafana (9 metrics, 10 alert rules) are all fully implemented.
- **Production Robustness:** The `/predict` endpoint handles stale data via a 2-stage backfill strategy; startup validates all artifacts before accepting traffic; drift detection runs after every hourly ingestion cycle.
- **Testing:** 47 automated test cases with a 100% pass rate, covering all critical paths — SQLite I/O, API contracts, model helper functions, and drift detection statistical logic.
- **Loose Coupling:** Streamlit and FastAPI are fully independent Docker containers communicating only via REST, with no shared state.

The primary deferred item is deep learning (LSTM / Transformer) due to hardware constraints. All pipeline infrastructure, preprocessing scripts, and DVC stages to support DL training and evaluation are already in place for Phase 2.

A Appendix A — Environment Variables

Table 23: All configurable environment variables (`.env`)

Variable	Required	Default / Description
`OPENAQ_API_KEY`	**Yes**	OpenAQ v3 API key (`explore.openaq.org`)
`OPENAQ_BASE_URL`	No	`https://api.openaq.org/v3`
`OPEN_METEO_ARCHIVE_URL`	No	`https://archive-api.open-meteo.com/v1/archive`
`MLFLOW_TRACKING_URI`	No	`file:///tmp/mlruns`
`MODEL_NAME`	No	Filename of `model.pkl`
`MODEL_STAGE`	No	`production`
`MODEL_PATH`	No	Path to `model.pkl`
`MODEL_METADATA_PATH`	No	Path to `model_metadata.json`
`FEATURE_COLUMNS_PATH`	No	Path to `feature_columns.json`
`TRAINED_CITIES_PATH`	No	Path to `trained_cities.json`
`CITY_ENCODER_PATH`	No	Path to `city_label_encoder.pkl`
`CITY_COORDS_PATH`	No	Path to `city_coords.json`
`DB_PATH`	No	`database/live_aqi_buffer.db`
`ENABLE_STARTUP_BACKFILL`	No	`false` — if `true`, backfills all cities on startup
`LIVE_HISTORY_HOURS`	No	`200` — hours of history to fetch per backfill

B Appendix B — params.yaml (Full)

```yaml
mode: all
city: Chennai

data:
  start_date: '2015-01-01'
  end_date:   '2020-12-31'
```

```yaml

preprocess:
  null_threshold: 0.8

training:
  model:            lgbm
  forecast_horizon: 24
  sample_frac:      1.0

lgbm:
  n_estimators:       200
  learning_rate:      0.05
  num_leaves:         31
  max_depth:          6
  min_child_samples:  50
  optuna:
    enabled:              false
    n_trials:             5
    random_seed:          42
    n_estimators_min:     100
    n_estimators_max:     300
    learning_rate_min:    0.01
    learning_rate_max:    0.2
    num_leaves_min:       20
    num_leaves_max:       100
    max_depth_min:        3
    max_depth_max:        8
    min_child_samples_min: 10
    min_child_samples_max: 100

xgboost:
  n_estimators:  500
  learning_rate: 0.05
  max_depth:     6
  subsample:     0.8
  optuna:
    enabled:          false
    n_trials:         10
    random_seed:      42
    n_estimators_min:  100
    n_estimators_max:  300
    learning_rate_min: 0.01
    learning_rate_max: 0.2
    max_depth_min:    3
    max_depth_max:    8
    subsample_min:    0.6
    subsample_max:    1.0
```

```yaml

lstm:
  seq_len:        24
  n_ahead:        24
  hidden_size:    128
  num_layers:     2
  dropout:        0.2
  batch_size:     512
  epochs:         50
  learning_rate:  0.001
  patience:       5

transformer:
  seq_len:        24
  n_ahead:        24
  d_model:        128
  nhead:          8
  num_layers:     3
  dropout:        0.1
  batch_size:     512
  epochs:         50
  learning_rate:  0.0001
  patience:       5
```

Listing 6: Full `params.yaml` — all pipeline and model parameters

C Appendix C — Backend Requirements

```
fastapi==0.115.0
uvicorn[standard]==0.30.6
pydantic==2.8.2
requests==2.32.3
pandas==2.2.3
numpy==2.1.3
scikit-learn==1.5.2
joblib==1.4.2
pyarrow==18.1.0
python-dotenv==1.0.1
prometheus-fastapi-instrumentator==7.0.0
lightgbm==4.3.0
httpx==0.27.0
scipy==1.14.1
prometheus-client==0.21.0
mlflow==3.11.1
```

```
mlflow-skinny==3.11.1
mlflow-tracing==3.11.1
```

Listing 7: backend/requirements.txt

D Appendix D — Quick-Start Commands

```
# 1. Clone and configure
git clone <your-repo-url>
cd AQIPredictor
cp .env.example .env
# Edit .env and set OPENAQ_API_KEY

# 2. Start main project services
docker-compose up --build

# Start Airflow separately
cd airflow && docker-compose up -d
# Visit http://localhost:8080 (user: airflow / pass: airflow)

# 3. Trigger first ingestion (manual)
curl -X POST http://localhost:8000/admin/sync

# 4. Get a prediction
curl "http://localhost:8000/predict?city=Chennai&location_id=3135"

# 5. Run the full test suite
cd backend && pytest tests/ -v --tb=short

# 6. Reproduce a specific experiment
git checkout lgbm-exp4
dvc repro train
mlflow ui  # visit http://localhost:5000

# 7. Generate DVC DAG image
dvc dag --dot | dot -Tpng -o figures/dvc_dag.png

# 8. Check pipeline status
dvc status
dvc dag
```

Listing 8: From zero to running system